Machine Learning

Lecture 04 — Questions Bank

Ensemble Methods · Bias/Variance · Bagging · Random Forests
Optimization (Gradient Descent) · Feature Mapping

MCQ · 12 Essay · 6 Numerical 30
مع الإجابات الصحيحة والشروحات

أسئلة الاختيار من متعدد — Multiple Choice Questions (MCQ)

1. What is the primary goal of ensemble methods in machine learning? high

ما هو الهدف الأساسي من أساليب التجميع (Ensemble Methods) في تعلم الآلة؟

A. Reduce model complexity

B. Combine multiple models to improve prediction accuracy and stability

C. Increase training speed

D. Reduce the size of the training dataset

✅ **الإجابة الصحيحة: Combine multiple models to improve prediction accuracy and stability (B)**

💡 ليه صح: أساليب التجميع بتجمع نماذج ضعيفة لتكوين نموذج قوي يحسن الدقة والاستقرار. الباقي غلط لأن تقليل التعقيد أو السرعة أو البيانات مش الهدف الأساسي.

📍 المرجع: Lecture 04 — Introduction / Ensemble Methods

2. Bias in machine learning corresponds to which problem? high

الانحياز (Bias) في تعلم الآلة بيقابل أي مشكلة؟

A. Over-fitting

B. High variance

C. Under-fitting

D. Data noise

✅ **الإجابة الصحيحة: Under-fitting (C)**

💡 ليه صح: الـ Bias بيقابل Under-fitting لأن النموذج البسيط جدًا مش قادر يلتقط اتجاه البيانات. الـ Over-fitting بيقابل الـ Variance مش الـ Bias.

📍 المرجع: Lecture 04 — Bias/Variance Section

3. Variance in machine learning corresponds to which problem? high

التباين (Variance) في تعلم الآلة بيقابل أي مشكلة؟

A. Under-fitting

B. Over-fitting

C. High bias

D. Model simplicity

✅ الإجابة الصحيحة: Over-fitting (B)

💡 ليه صح: الـ Variance بيقابل Over-fitting لأن النموذج بيبقى حساس جدًا للتغييرات في بيانات التدريب وبيتعلم من الـ noise. الـ Under-fitting بيقابل الـ Bias.

📍 المرجع: Lecture 04 — Variance Section

4. An overly simple model (e.g., KNN with large k) typically has: high

النموذج البسيط جدًا (زي KNN بقيمة k كبيرة) عادة بيكون عنده:

A. High bias and high variance

B. Low bias and high variance

C. High bias and low variance

D. Low bias and low variance

✅ الإجابة الصحيحة: High bias and low variance (C)

💡 ليه صح: النموذج البسيط عنده High bias (مش قادر يلتقط الهيكل) و Low variance (تقديرات مستقرة). ده عكس النموذج المعقد اللي عنده Low bias و High variance.

📍 المرجع: Lecture 04 — Overly Simple Model Slide

5. An overly complex model (e.g., KNN with k = 1) typically has: high

النموذج المعقد جدًا (زي KNN بقيمة k = 1) عادة بيكون عنده:

A. High bias and high variance

B. High bias and low variance

C. Low bias and low variance

D. Low bias and high variance

✅ الإجابة الصحيحة: Low bias and high variance (D)

💡 ليه صح: النموذج المعقد عنده Low bias (بيتعلم كل الهيكل) و High variance (بيناسب غرائب البيانات). ده عكس النموذج البسيط.

📍 المرجع: Lecture 04 — Overly Complex Model Slide

6. What is a bootstrap sample in bagging? high

إيه هو الـ bootstrap sample في الـ Bagging؟

A. A subset sampled without replacement

B. The entire training dataset

C. A subset of features only

D. A random subset of the data sampled with replacement

✅ الإجابة الصحيحة: A random subset of the data sampled with replacement (D)

💡 ليه صح: الـ bootstrap sample عينة عشوائية مع الاستبدال (with replacement)، يعني نفس النقطة ممكن تختار أكتر من مرة. مش subset بدون استبدال ولا الـ dataset كله.

📍 المرجع: Lecture 04 — Bagging: The Idea

7. What does bagging primarily reduce? high

الـ Bagging بقلل أساسًا إيه؟

A. Variance

B. Bias

C. Both bias and variance equally

D. Neither bias nor variance

✅ الإجابة الصحيحة: Variance (A)

💡 ليه صح: الـ Bagging بقلل الـ Variance عن طريق المتوسط بين تنبؤات مستقلة. الـ Bias بيفضل زي ما هو لأن المتوسط ليه نفس الـ expectation.

📍 المرجع: Lecture 04 — Bagging Properties

8. What is the effect of bagging on bias (for squared error)? high

إيه تأثير الـ Bagging على الـ Bias (في حالة الخطأ التربيعي)؟

A. It increases bias

B. It decreases bias significantly

C. It makes bias zero

D. Bias remains unchanged

✅ الإجابة الصحيحة: Bias remains unchanged (D)

💡 ليه صح: الـ Bias بيفضل unchanged لأن المتوسط للتنبؤات ليه نفس الـ expectation. ده من محدوديات الـ Bagging: مش بقلل الـ Bias في حالة squared error.

📍 المرجع: Lecture 04 — Bagging Summary / Limitations

9. How are predictions aggregated in bagging? medium

إزاي بتتجمع التنبؤات في الـ Bagging؟

A. Only by voting

B. By voting, averaging, or weighting

C. Only by averaging

D. By taking the maximum prediction

✅ الإجابة الصحيحة: By voting, averaging, or weighting (B)

💡 ليه صح: التنبؤات بتتجمع عن طريق voting أو averaging أو weighting. محدش منهم الطريقة الوحيدة، والـ maximum مش طريقة صحيحة.

📍 المرجع: Lecture 04 — Ensemble Methods Overview

10. How does Random Forest reduce correlation between bagged decision trees? high

إزاي الـ Random Forest بقلل الارتباط بين أشجار القرار المجمعة؟

A. By using fewer trees in the forest

B. By using smaller training datasets

C. By choosing a random subset of features at each node

D. By limiting the depth of every tree

✅ الإجابة الصحيحة: By choosing a random subset of features at each node (C)

💡 ليه صح: الـ Random Forest بيختار subset عشوائي من الميزات عند كل node. ده بيقول الـ correlation بين الأشجار، على عكس تقليل عدد الأشجار أو العمق.

📍 المرجع: Lecture 04 — Random Forests (Trick Section)

11. The "wisdom of the crowd" concept in Random Forests means: medium

مفهوم "حكمة الجمهور" في الـ Random Forests معناه:

A. A large group of experts is collectively smarter than individual experts

B. A single expert is always better than a group

C. Crowd voting always produces wrong answers

D. Each individual tree must be perfect

✅ الإجابة الصحيحة: A large group of experts is collectively smarter than individual experts (A)

💡 ليه صح: مجموعة كبيرة من الخبراء بتكون أذكى جماعيًا من الأفراد. ده المبدأ الأساسي وراء الـ Random Forest. الباقي غلط.

📍 المرجع: Lecture 04 — Random Forests Concepts

12. In optimization, a direct solution to minimize J(w) is obtained by: high

في الـ Optimization، الحل المباشر لتقليل J(w) بيتحصل بإيه؟

A. Iteratively updating the weights

B. Randomly searching the weight space

C. Setting the gradient to zero and solving in closed form

D. Using a small learning rate

✅ الإجابة الصحيحة: Setting the gradient to zero and solving in closed form (C)

💡 ليه صح: الحل المباشر بيساوي الـ gradient للصفر ويحل بشكل مغلق (closed form). الـ Iterative updating ده طريقة Gradient Descent.

📍 المرجع: Lecture 04 — Solving the Minimization Problem

13. Gradient descent is what type of algorithm? high

الـ Gradient Descent هو خوارزمية من أي نوع؟

A. Iterative

B. Direct / closed-form

C. Random search

D. Brute-force enumeration

✅ الإجابة الصحيحة: Iterative (A)

💡 ليه صح: الـ Gradient Descent خوارزمية Iterative بنطبق فيها update rule بشكل متكرر لحد ما نحقق معيار معين.

📍 المرجع: Lecture 04 — Iterative Solution: Gradient Descent

14. The gradient descent update rule is: high

قاعدة تحديث الـ Gradient Descent هي:

A. $w_j \rightarrow w_j + \alpha \, \partial J/\partial w_j$

B. $w_j \rightarrow \alpha \, \partial J/\partial w_j$

C. $w_j \rightarrow w_j - \partial J/\partial w_j$

D. $w_j \rightarrow w_j - \alpha \, \partial J/\partial w_j$

✅ الإجابة الصحيحة: $w_j \rightarrow w_j - \alpha \, \partial J/\partial w_j$ (D)

💡 ليه صح: بنقص α ضرب المشتقة من الوزن. الخيار الأول غلط لأنه بيزود بدل ما ينقص، والباقي ناقص α أو المشتقة.

📍 المرجع: Lecture 04 — Deriving Update Rule

15. In the gradient descent update rule, α represents: high

في قاعدة تحديث الـ α، Gradient Descent بتمثل:

A. The gradient value

B. The learning rate

C. The weight parameter

D. The cost function

✅ الإجابة الصحيحة: The learning rate (B)

💡 ليه صح: α هي الـ learning rate أو step size، قيمة موجبة صغيرة بتحدد حجم الخطوة.

📍 المرجع: Lecture 04 — Setting Learning Rate

16. If ∂J/∂wj > 0, to decrease J we should: medium

لو ∂J/∂wj > 0، عشان نقلل J لازم:

A. Increase wj

B. Keep wj unchanged

C. Decrease wj

D. Set wj to zero

✅ الإجابة الصحيحة: Decrease wj (C)

💡 ليه صح: لما المشتقة موجبة، تقليل J بيحتاج تقليل wj. لو زودنا wj هيزود J.

📍 المرجع: Lecture 04 — Deriving Update Rule

17. Typical values of the learning rate α are: medium

القيم النموذجية لـ learning rate α هي:

A. 1.0 or 10.0

B. 100 or 1000

C. Negative values like −0.5

D. 0.01 or 0.0001

✅ الإجابة الصحيحة: 0.01 or 0.0001 (D)

💡 ليه صح: القيم النموذجية صغيرة زي 0.01 أو 0.0001. القيم الكبيرة بتسبب divergence، والسالب غلط لأن α لازم يكون موجب.

📍 المرجع: Lecture 04 — Setting Learning Rate

18. Why is gradient descent preferred over the direct solution for high-dimensional regression? high

ليه الـ Gradient Descent مفضل على الحل المباشر في الانحدار عالي الأبعاد؟

A. Direct solution requires matrix inversion which is O(D^3)

B. Gradient descent is always more accurate

C. Gradient descent always converges to the global minimum

D. Direct solution cannot work for regression

✅ الإجابة الصحيحة: Direct solution requires matrix inversion which is O(D^3) (A)

💡 ليه صح: الحل المباشر بيحتاج matrix inversion بتكلفة $O(D^3)$، وده مكلف في الأبعاد العالية. الـ GD update بتكلف O(ND) أو أقل مع SGD.

📍 المرجع: Lecture 04 — Why Use Gradient Descent?

19. The cost of a single gradient descent update (or stochastic GD) is approximately:

high

تكلفة تحديث واحد في الـ Gradient Descent (أو SGD) تقريبًا:

A. O(D^3)

B. O(ND) or less with stochastic gradient descent

C. O(N^2)

D. O(D^2)

✅ الإجابة الصحيحة: O(ND) or less with stochastic gradient descent (B)

💡 ليه صح: تكلفة تحديث GD هي O(ND) أو أقل مع SGD. O(D³) دي تكلفة الحل المباشر.

📍 المرجع: Lecture 04 — Why Use Gradient Descent?

20. Feature mapping ψ(x) transforms the input from: high

الـ Feature Mapping ψ(x) بتحول المدخلات من:

A. A lower-dimensional space to a higher-dimensional space

B. A higher-dimensional space to a lower-dimensional space

C. The same dimensional space to itself

D. Categorical to numerical data only

✅ الإجابة الصحيحة: A lower-dimensional space to a higher-dimensional space (A)

💡 ليه صح: الـ Feature Mapping بتحول من فضاء أقل أبعاد لفضاء أعلى أبعاد، حيث البيانات تقدر تتحلل أو تتصنف بسهولة أكبر.

📍 المرجع: Lecture 04 — Feature Mapping

21. For polynomial feature mapping of degree M, the feature vector is: high

للـ Polynomial Feature Mapping من الدرجة M، الـ feature vector هو:

A. ψ(x) = [x, x^2, ..., x^M]

B. ψ(x) = [1, x, ..., x^(M−1)]

C. ψ(x) = [1, x, x^2, ..., x^M]

D. ψ(x) = [x, x^2, ..., x^(M+1)]

✅ الإجابة الصحيحة: ψ(x) = [1, x, x^2, ..., x^M] (C)

💡 ليه صح: الـ feature vector الصحيح بيبدأ بـ 1 (للـ bias term w0) وينتهي بـ x^M. الخيارات التانية ناقصة الـ 1 أو بتعدى M.

📍 المرجع: Lecture 04 — Polynomial Feature Mapping

22. Which polynomial degree M shows under-fitting according to the lecture? high

أي درجة كثيرة M بتظهر Under-fitting حسب المحاضرة؟

A. M = 0

B. M = 3

C. M = 9

D. M = 12

✅ الإجابة الصحيحة: M = 0 (A)

💡 ليه صح: M = 0 (y = w0) بيتسبب في Under-fitting لأنه مش قادر يلتقط اتجاه البيانات. M = 3 نموذج جيد، و M = 9 بيعمل over-fitting.

📍 المرجع: Lecture 04 — Model Complexity and Generalization

23. Which polynomial degree M shows over-fitting according to the lecture? high

أي درجة كثيرة M بتظهر Over-fitting حسب المحاضرة؟

A. M = 0

B. M = 1

C. M = 3

D. M = 9

✅ **الإجابة الصحيحة: M = 9 (D)**

💡 ليه صح: M = 9 (نموذج معقد جدًا) بيتسبب في Over-fitting، بيمثل بيانات التدريب بشكل مثالي لكن بيفشل في بيانات اختبار جديدة.

📍 المرجع: Lecture 04 — Model Complexity and Generalization

24. Which polynomial degree M gives a good model that generalizes well? high

أي درجة كثيرة M بتدّي نموذج جيد بيعمم كويس؟

A. M = 0

B. M = 3

C. M = 9

D. M = 12

✅ **الإجابة الصحيحة: M = 3 (B)**

💡 ليه صح: M = 3 بيدي نموذج جيد بـ test error صغير وبيعمم كويس. M = 0 under-fitting و M = 9 over-fitting.

📍 المرجع: Lecture 04 — Model Complexity and Generalization

25. The Random Subspace Method is also known as: medium

الـ Random Subspace Method معروف كمان بإسم:

A. Feature Bagging

B. Data Bagging

C. Boosting

D. Stacking

✅ الإجابة الصحيحة: Feature Bagging (A)

💡 ليه صح: الـ Random Subspace Method معروف كـ Feature Bagging لأنه بيشتغل على subset عشوائي من الميزات. Boosting و Stacking أساليب تانية خالص.

📍 المرجع: Lecture 04 — Random Forest (Bagging + Random Subspace)

26. Random Forests can be used for which type of tasks? medium

الـ Random Forests ممكن يُستخدم لأي نوع من المهام؟

A. Classification only

B. Both classification and regression

C. Regression only

D. Clustering only

✅ الإجابة الصحيحة: Both classification and regression (B)

💡 ليه صح: الـ Random Forest خوارزمية supervised بتشتغل لـ Classification و Regression. مش للـ Clustering لأنه supervised.

📍 المرجع: Lecture 04 — Random Forests

27. Which of the following is NOT a way to reduce high bias? high

أي من الخيارات دي مش طريقة لتقليل الـ Bias العالي؟

A. Use a more complex model

B. Increase the number of features

C. Increase the size of the training data

D. Reduce the number of features

✅ الإجابة الصحيحة: Reduce the number of features (D)

💡 ليه صح: تقليل عدد الميزات بيقلل التعقيد وبالتالي يزيد الـ Bias. الطرق الصحيحة: نموذج أعقد، ميزات أكتر، أو بيانات تدريب أكتر.

📍 المرجع: Lecture 04 — Ways to reduce high bias

28. Which of the following is a valid way to reduce variance? high

أي من الخيارات دي طريقة صحيحة لتقليل الـ Variance؟

A. Use a more complex model

B. Add more features to the model

C. Use cross-validation

D. Increase the polynomial degree M

✅ الإجابة الصحيحة: Use cross-validation (C)

💡 ليه صح: الـ Cross-validation طريقة صحيحة لتقليل Variance عن طريق تقسيم البيانات عدة مرات. باقي الخيارات بتزيد التعقيد و بالتالي تزيد الـ Variance.

📍 المرجع: Lecture 04 — Ways to Reduce the Variance

29. A key limitation of bagging is: high

أي من الخيارات دي محدودية أساسية للـ Bagging؟

A. It increases the variance of the model

B. It cannot be used for classification tasks

C. It does not reduce bias in case of squared error

D. It always overfits the training data

✅ الإجابة الصحيحة: It does not reduce bias in case of squared error (C)

💡 ليه صح: الـ Bagging مش بقلل الـ Bias في حالة squared error. هو بقلل الـ Variance مش بيزيده، وبيشتغل للـ classification، ومش بيoperfit.

📍 المرجع: Lecture 04 — Bagging Summary / Limitations

30. If the m datasets in bagging are NOT independent, what happens? high

لو الـ m datasets في الـ Bagging مش مستقلة، إيه اللي بيحصل؟

A. We get the full 1/m variance reduction

B. Variance becomes exactly zero

C. Bias increases dramatically

D. We do not get the full 1/m variance reduction

✅ الإجابة الصحيحة: We do not get the full 1/m variance reduction (D)

💡 ليه صح: لو الـ datasets مش مستقلة، مبنحصلش على تخفيض الـ Variance الكامل بمقدار 1/m. عشان كده بنستخدم Random Forest عشان نقلل الـ correlation.

📍 المرجع: Lecture 04 — Bagging Properties

الأسئلة المقالية — Essay Questions

1. Define Bias and Variance in machine learning, and explain what each one corresponds to. high

عرّف الـ Bias والـ Variance في تعلم الآلة، ووضّح كل واحد فيهم بيقابل إيه.

Bias is the error due to wrong assumptions about the data (e.g., assuming the data is linear when it follows a complex function).
It measures how wrong the expected prediction is, and corresponds to under-fitting (a high-bias model cannot capture the dataset trend).
Variance is the variability of the model — how sensitive it is to a different subset of the training data.
It corresponds to over-fitting, where the model fits the training data too closely and performs poorly on new, unseen data.

📍 Reference: Lecture 04 — Bias/Variance Section

2. Explain the bias-variance tradeoff using KNN with different values of k. high

اشرح المقايضة بين الـ Bias والـ Variance باستخدام KNN بقيم k مختلفة.

KNN with large k is an overly simple model: it has high bias (cannot capture the structure in the data) and low variance (stable estimates because it averages many neighbors).
KNN with k = 1 is an overly complex model: it has low bias (learns all the relevant structure) and high variance (fits the quirks of the particular sampled data).
The tradeoff: as model complexity increases, bias decreases but variance increases. The goal is to find a balance that minimizes total error on unseen data.

📍 Reference: Lecture 04 — Overly Simple/Complex Model Slides

3. Compare Bagging and Random Forests. What do they share and how do they differ?

high

قارن بين الـ Bagging والـ Random Forests. إيه المشترك وإيه الفرق؟

Similarities: Both are ensemble methods that train multiple models on random subsets of data (bootstrapped) and aggregate predictions (voting/averaging). Both aim to reduce variance.
Differences: Bagging uses the same algorithm on different bootstrapped datasets, while Random Forest specifically uses decision trees and adds an extra layer of randomness by selecting a random subset of features at each node (Random Subspace Method / Feature Bagging).
This extra randomness in Random Forest reduces the correlation between trees, which gives a stronger variance reduction than plain bagging.
Random Forest is described as the best black-box ML algorithm — it works well with no tuning and is widely used in Kaggle competitions.

📍 Reference: Lecture 04 — Random Forests & Bagging Properties

4. Explain how bagging reduces variance. Why does it leave bias unchanged?

high

اشرح إزاي الـ Bagging بقلل الـ Variance. وليه بسيب الـ Bias زي ما هو؟

Bagging trains m models independently on m different bootstrap samples of the data.
Variance is reduced because we are averaging over (approximately) independent predictions — averaging reduces variability.
Bias remains unchanged because the averaged prediction has the same expectation as the individual predictions — taking the average does not change what the model is fundamentally able to learn.
This is why bagging is most useful on high-variance, low-bias models (like deep decision trees).

📍 Reference: Lecture 04 — Bagging (effect on bias and variance)

5. List and briefly explain the ways to reduce high bias in machine learning. high

اذكر واشرح بإختصار طرق تقليل الـ Bias العالي في تعلم الآلة.

1) Use a more complex model — e.g., add hidden layers in a deep neural network, or use polynomial regression for non-linear datasets.
2) Increase the number of features — adding more features increases model complexity and helps capture underlying patterns in the data.
3) Increase the size of the training data — providing the model with more examples to learn from can help reduce bias.

📍 Reference: Lecture 04 — Ways to reduce high bias

6. List and briefly explain the ways to reduce variance in machine learning. high

اذكر واشرح بإختصار طرق تقليل الـ Variance في تعلم الآلة.

1) Cross-validation — split the data into training and testing sets multiple times.
2) Feature selection — choosing only the relevant features decreases the model's complexity.
3) Ensemble methods — combine multiple models to improve generalization (e.g., Bagging, Boosting).
4) Simplifying the model — reduce complexity such as decreasing the number of parameters or layers in a neural network.

📍 Reference: Lecture 04 — Ways to Reduce the Variance

7. Explain gradient descent and derive its update rule. What is the role of the learning rate α? high

اشرح الـ Gradient Descent واشتق قاعدة التحديث. إيه دور الـ learning rate α؟

Gradient descent is an iterative optimization algorithm used to minimize a cost function J(w).
Key observation: if ∂J/∂wj > 0, decreasing J requires decreasing wj; if ∂J/∂wj < 0, decreasing J requires increasing wj.
Update rule: wj → wj − α * (∂J / ∂wj). This update always decreases the cost function for small enough α (unless ∂J/∂wj = 0).
The learning rate α > 0 controls the step size. Larger α means faster weight changes; typical values are small (e.g., 0.01 or 0.0001). Too large α can cause divergence; too small α makes convergence slow.

📍 Reference: Lecture 04 — Iterative Solution: Gradient Descent & Deriving Update Rule

8. Compare direct solution and iterative (gradient descent) solution for minimizing the cost function J(w). high

قارن بين الحل المباشر والحل التكراري (Gradient Descent) لتقليل دالة التكلفة J(w).

Direct solution: set the gradient ∇w J(w) = 0 and solve in closed form to directly find provably optimal parameters.
Iterative solution (Gradient Descent): repeatedly apply an update rule that gradually moves the weights toward the solution.
Why use gradient descent over direct solution:
1) Applicable to a much broader set of models (many problems have no closed-form solution).
2) Easier to implement than direct solutions.
3) More efficient for regression in high-dimensional space — direct solution for linear regression $(X^\top X)^{-1} X^\top t$ requires matrix inversion which is O(D^3), while a GD update costs only O(ND) or less with stochastic GD.

📍 Reference: Lecture 04 — Solving the Minimization Problem & Why Use GD

9. Explain feature mapping. Why is it used, and how does it relate to linear regression? high

اشرح الـ Feature Mapping. ليه بتُستخدم، وإزاي بترتبط بالـ Linear Regression؟

Feature mapping transforms input features from a lower-dimensional space to a higher-dimensional space using a function ψ(x): R^D → R^d, where the data can be more easily analyzed or classified.
It allows linear regression to model non-linear relationships — by mapping the input features to another space and then treating the mapped features as the input of a linear regression procedure.
Example (Polynomial feature mapping): for a degree-M polynomial, ψ(x) = [1, x, x^2, ..., x^M]⊤.
The model y = ψ(x)⊤w is still linear in w0, w1, ..., so we can use linear regression to find w even though the relationship with x is non-linear.

📍 Reference: Lecture 04 — Feature Mapping & Polynomial Feature Mapping

10. Discuss model complexity and generalization using polynomial feature mapping with different values of M. high

ناقش تعقيد النموذج والتعميم باستخدام Polynomial Feature Mapping بقيم M مختلفة.

Using polynomial feature mapping of degree M:
M = 0 (y = w0): under-fitting — the model is too simple and does not fit the data well.
M = 1 (y = w0 + w1*x): still a simple linear fit.
M = 3 (y = w0 + w1*x + w2*x^2 + w3*x^3): good model — small test error, generalizes well.
M = 9 (y = w0 + ... + w9*x^9): over-fitting — the model is too complex, fits the training data perfectly but fails on new data.
Conclusion: as M (complexity) increases, training error decreases, but test error first decreases then increases. The goal is to choose M that minimizes test error, not training error.

📍 Reference: Lecture 04 — Polynomial Feature Mapping (M=0,1,3,9) & Model Complexity

11. Explain the Random Subspace Method (Feature Bagging) and how it is used in Random Forests. medium

اشرح الـ Random Subspace Method (Feature Bagging) وإزاي بيُستخدم في الـ Random Forests.

The Random Subspace Method (also called Feature Bagging) fits N different decision trees by constraining each one to operate on a random subset of features.
In Random Forests, both techniques are combined:
1) Bagging — take random subsets of data points (with replacement) to create N smaller datasets and fit a tree on each subset, giving low variance.
2) Random Subspace Method — at each node of each tree, choose a random subset of features and consider splits on these features only.
This extra randomness at the node level reduces correlation between the bagged trees, which gives a stronger variance reduction than plain bagging alone. This is the trick that makes Random Forests work so well.

📍 Reference: Lecture 04 — Random Forest (Bagging + Random Subspace)

12. List and briefly explain at least six techniques for feature mapping mentioned in the lecture. medium

اذكر واشرح بإختصار ست تقنيات على الأقل للـ Feature Mapping مذكورة في المحاضرة.

1) Feature extraction — transform input data into a new set of features that capture the most important information (e.g., extracting edges, corners, textures from images).
2) Feature transformation — apply mathematical functions to transform input into new features (e.g., bag-of-words representation in text classification).
3) Feature selection — select a subset of the available features that are most relevant to the task.
4) Feature scaling — scale features so they have similar ranges and are on the same scale (important for distance-based algorithms).
5) Feature engineering — create new features from existing ones (e.g., ratio of transaction amount to average transaction amount in fraud detection).
6) Dimensionality reduction — reduce the number of features while retaining the most important information; helps reduce overfitting and improve computational efficiency.
7) Embeddings — vector representations that capture semantic meaning (e.g., word2vec, GloVe for words in NLP).
8) Data augmentation — create new examples by applying transformations to existing data (e.g., rotating, flipping, cropping images).
9) Domain knowledge — use expert knowledge about the domain to guide feature selection/engineering.

📍 Reference: Lecture 04 — Techniques for feature mapping

الأسئلة الحسابية — Numerical Questions

1. (Numerical) For a polynomial feature mapping of degree M = 3 and an input x = 2, write the full feature vector ψ(x). high

(سؤال حسابي) لـ Polynomial Feature Mapping من الدرجة M = 3 ومدخل x = 2، اكتب الـ feature vector ψ(x) كامل.

Given: M = 3, x = 2.
Formula: ψ(x) = [1, x, x^2, x^3]⊤
Step 1: 1 = 1
Step 2: x = 2
Step 3: x^2 = 2^2 = 4
Step 4: x^3 = 2^3 = 8
Result: ψ(2) = [1, 2, 4, 8]⊤
Note: the feature vector has M + 1 = 4 components because it starts from x^0 = 1.

📍 Reference: Lecture 04 — Polynomial Feature Mapping

2. (Numerical) In gradient descent, the current weight is wj = 2.0, the partial derivative ∂J/∂wj = 0.5, and the learning rate α = 0.1. Compute the new value of wj after one update. high

(سؤال حسابي) في الـ Gradient Descent، الوزن الحالي wj = 2.0، المشتقة الجزئية ∂J/∂wj = 0.5، ومعدل التعلم α = 0.1. احسب القيمة الجديدة لـ wj بعد تحديث واحد.

Given: wj = 2.0, ∂J/∂wj = 0.5, α = 0.1
Update rule: wj_new = wj − α * (∂J / ∂wj)
Step 1: α * (∂J / ∂wj) = 0.1 * 0.5 = 0.05
Step 2: wj_new = 2.0 − 0.05 = 1.95
Result: wj_new = 1.95
Interpretation: because ∂J/∂wj > 0, the update correctly decreases wj to reduce J.

📍 Reference: Lecture 04 — Deriving Update Rule & Setting Learning Rate

3. (Numerical) For a polynomial feature mapping of degree M = 9, how many features (components) are in the feature vector ψ(x)? Justify your answer. high

(سؤال حسابي) لـ Polynomial Feature Mapping من الدرجة M = 9، كام ميزة (عنصر) في الـ feature vector ψ(x)؟ برر إجابتك.

Formula: ψ(x) = [1, x, x^2, ..., x^M]⊤
The vector contains the powers of x from x^0 (= 1) up to x^M.
Number of components = M + 1 (because we start counting from 0).
For M = 9: number of features = 9 + 1 = 10.
The features are: [1, x, x^2, x^3, x^4, x^5, x^6, x^7, x^8, x^9].
Conclusion: a degree-9 polynomial fit requires estimating 10 weights (w0 through w9).

📍 Reference: Lecture 04 — Polynomial Feature Mapping (M=9 case)

4. (Numerical) Compare the cost of the direct solution versus one gradient descent update for a regression problem with D = 100 features and N = 1000 training examples. high

(سؤال حسابي) قارن تكلفة الحل المباشر مقابل تحديث واحد في الـ Gradient Descent لمشكلة regression فيها D = 100 ميزة و N = 1000 مثال تدريب.

Direct solution cost (matrix inversion): O(D^3) = O(100^3) = 1,000,000 operations.
Gradient descent update cost: O(N*D) = O(1000 * 100) = 100,000 operations per update.
Comparison: GD update is about 10x cheaper than the direct solution per iteration.
Conclusion: For high-dimensional regression (large D), gradient descent is more efficient than the direct solution, which requires expensive matrix inversion. With stochastic gradient descent (SGD), the per-iteration cost can be even lower (O(D) per example).

📍 Reference: Lecture 04 — Why Use Gradient Descent?

5. (Numerical) Write the polynomial equation and the corresponding feature vector for the case M = 1, and explain what kind of model it represents. medium

(سؤال حسابي) اكتب معادلة الكثيرة والـ feature vector المقابل لحالة M = 1، واشرح نوع النموذج اللي بتمثله.

Polynomial equation for M = 1: y = w0 + w1*x
Feature vector: ψ(x) = [1, x]⊤
Number of features = M + 1 = 2.
Interpretation: M = 1 represents simple linear regression — a straight line that fits the data with a single slope (w1) and intercept (w0).
Note from the lecture: M = 1 is still a simple model. If the true relationship is non-linear, M = 1 (like M = 0) will under-fit the data. M = 3 is needed for a good fit on the lecture's example dataset.

📍 Reference: Lecture 04 — Polynomial Feature Mapping with M = 1

6. (Numerical) In the bagging example from the lecture, m = 3 datasets are created by sampling with replacement, each containing n = 7 examples. If the original dataset D has 7 examples, what is the total number of model predictions made and how is the final prediction obtained? high

(سؤال حسابي) في مثال الـ Bagging من المحاضرة، m = 3 datasets بتتخلق بأخذ عينات مع استبدال، كل واحد فيهم n = 7 أمثلة. لو الـ dataset الأصلي D فيه 7 أمثلة، إيه العدد الكلي للتنبؤات اللي بتتعمل وإزاي بيتحصل على التنبؤ النهائي؟

Given: m = 3 datasets, each with n = 7 examples, sampled with replacement from D (which has 7 examples).
Step 1 — Training: Train one model on each of the m = 3 bootstrap datasets, producing 3 trained models.
Step 2 — Prediction: For each test sample, each of the 3 models produces a prediction yi (i = 1, 2, 3). Total = 3 predictions per test sample.
Step 3 — Aggregation: Average the 3 predictions: y_final = (y1 + y2 + y3) / 3.
Conclusion: 3 model predictions are made per test sample, and the final prediction is the average of these 3 predictions.
Note: because sampling is with replacement, some original examples may appear multiple times in the same bootstrap dataset, and some may not appear at all.

📍 Reference: Lecture 04 — Bagging Example 1/2 & 2/2